



Project

Analyzing Databases

AIC - Data Analytics Specialization V2

Project Overview

Objective

As a data analyst, your team works across different data systems: structured business data in a SQL database, and flexible user behavior data in a NoSQL MongoDB.

Your goal is to connect to both databases using R in Google Colab, perform meaningful queries, and turn the results into insights that could guide business decisions.

Databases

You will engage with the two databases introduced during the module - the classicmodels (relational) and mfliX (non-relational) databases. You can access the classicmodels database [here](#), and connect to the mfliX database within your code.

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The classic models database schema consists of the following tables:

- ❑ **Customers:** stores customer's data.
- ❑ **Products:** stores a list of scale model cars.
- ❑ **ProductLines:** stores a list of product line categories.
- ❑ **Orders:** stores sales orders placed by customers.
- ❑ **OrderDetails:** stores sales order line items for each sales order.
- ❑ **Payments:** stores payments made by customers based on their accounts.
- ❑ **Employees:** stores all employee information as well as the organization structure such as who reports to whom.
- ❑ **Offices:** stores sales office data.

For the mflix database, we will focus on the collection named “movies” only.

Expected Work

From the SQL database:

Choose **two** business-relevant questions to investigate.

Your queries should go beyond one-table lookups. This means joining multiple tables, using filters or conditions, and calculating something insightful, such as totals, averages, counts, or rankings.

Think in terms of what a stakeholder would want to know, not just “what’s in the table,” but “what does the data tell us?”

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Expected Work

Example question types (you must write your own):

- ❑ Which products have generated the highest revenue?
- ❑ Are some employees consistently closing more or higher-value orders?
- ❑ Which offices are linked to the most late shipments?

From the mflix database:

Choose **two insight-focused questions** about movies or users.

Your queries should involve **filtering, grouping, and aggregating** across fields. Consider using **MongoDB's aggregation pipeline** to answer more complex or trend-based questions.

Aim to uncover patterns, rankings, or comparisons that would be useful for decision-making, not just pull raw data from one collection.

Example question types:

- ❑ What genres have the highest average ratings?
- ❑ Which directors consistently produce top-rated films?
- ❑ Are there regions where people watch longer movies?

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Deliverables (Read Carefully!)

Submit a **completed Google Colab notebook** that includes:

- ❑ Your **code in R language** for both SQL and MongoDB queries
- ❑ Clear section headers separating:
 - ❑ SQL-based analysis
 - ❑ MongoDB (NoSQL) analysis
- ❑ All necessary **data loading or connection setup code**
- ❑ At least **two well-written queries** for each database
- ❑ Output summaries or key statistics (you don't need to display full tables)
- ❑ A brief explanation (in markdown cells) after each query:
 - ❑ What the result shows
 - ❑ Why it matters to the business or product team

Your notebook should run from start to finish without errors, and be readable for someone reviewing your work like a stakeholder or teammate.

Important Notes:

- ❑ There's **no one “correct”** way to approach this challenge. Use your creativity and critical thinking to guide your decisions.
- ❑ Any submission flagged for plagiarism, unauthorized collaboration, or AI-generated answers **will either receive a penalty or be rejected entirely.**

The Extra Mile *(Optional – Earn extra points)*

Each challenge gives you a chance to go beyond what's required. We call this **The Extra Mile**.

For this module, your Extra Mile task is to complete [this short quiz](#) that tests your real-world understanding of how databases work, from relational models and ER diagrams to choosing between SQL and NoSQL.

Submit your answers by the challenge deadline and you will earn **extra points**.